

The elementary-symmetric candidate on a Fermat fourfold is an ambient complete intersection

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Abstract

Let $X_{3d} \subset \mathbf{P}_{\mathbf{C}}^5$ be the Fermat fourfold and let W_d be the surface proposed in Question 1 of da Silva's 2021 notes, defined by the first three elementary symmetric polynomials in $x_0^d, \dots, x_5^d$. Newton's identity proves scheme-theoretically that W_d is the intersection on X_{3d} of two ambient divisors. Consequently

$$[W_d] = 2d^2 H^2 \in \mathrm{CH}^2(X_{3d}),$$

so its primitive cohomology class and every primitive Fermat-character projection vanish. This gives a negative answer for the proposed candidate, without deciding algebraicity of the exceptional characters. We also record the independent fact that any cycle class invariant under coordinatewise d -th roots of unity can occur only in character sectors whose coordinates are all divisible by d .

1 The candidate

Let k be a field in which 3 is invertible, let $d \geq 1$, and put

$$S = k[x_0, \dots, x_5], \quad y_i = x_i^d, \quad F_{3d} = \sum_{i=0}^5 x_i^{3d}.$$

Write $e_j = e_j(y_0, \dots, y_5)$ for the elementary symmetric polynomials. Define

$$X_{3d} = V(F_{3d}) \subset \mathbf{P}_k^5, \quad W_d = V(e_1, e_2, e_3) \subset \mathbf{P}_k^5.$$

This is the candidate denoted W in Question 1 of [1]. The condition $3 \nmid d$ used there is not needed below.

2 Scheme structure and cycle class

Theorem 2.1. *If 3 is invertible in k , then*

$$(e_1, e_2, e_3) = (e_1, e_2, F_{3d}) \subset S.$$

This ideal is a complete-intersection ideal of height three. Hence

$$W_d = V_{X_{3d}}(e_1, e_2)$$

scheme-theoretically, and W_d is a pure two-dimensional complete intersection of type $(d, 2d, 3d)$ in $\mathbf{P}^5$.

Proof. Newton's third identity is

$$\sum_{i=0}^5 y_i^3 = e_1^3 - 3e_1e_2 + 3e_3.$$

After $y_i = x_i^d$, this becomes

$$F_{3d} = e_1^3 - 3e_1e_2 + 3e_3. \quad (1)$$

It gives both inclusions between the displayed ideals, the reverse inclusion using division by 3.

For completeness, let $A = k[y_0, \dots, y_5]$. The elementary symmetric functions are algebraically independent, and A is integral over $k[e_1, \dots, e_6]$. Thus (e_1, e_2, e_3) has height three in the Cohen–Macaulay ring A , and its generators form a regular sequence. The map

$$A \longrightarrow S, \quad y_i \longmapsto x_i^d,$$

is finite free, with basis $\prod_i x_i^{r_i}$ for $0 \leq r_i < d$. Flat base change preserves the regular sequence. Equation (1) then permits replacing e_3 by F_{3d} . The claimed height, purity, and scheme-theoretic description follow. $\square$

Now take $k = \mathbf{C}$ and let $H = c_1(\mathcal{O}_{X_{3d}}(1))$.

Corollary 2.2. *In the Chow group of the smooth Fermat fourfold,*

$$[W_d] = 2d^2H^2 \in \mathrm{CH}^2(X_{3d}), \quad \deg W_d = 6d^3.$$

Its cohomology class has zero primitive component and therefore zero projection to every primitive Fermat-character eigenspace.

Proof. By Theorem 2.1, W_d is the proper scheme-theoretic intersection on X_{3d} of Cartier divisors of classes dH and $2dH$. Refined intersection, including scheme multiplicities [2], gives

$$[W_d] = (dH)(2dH) = 2d^2H^2.$$

Since $\int_{X_{3d}} H^4 = 3d$, its degree is $6d^3$. The cycle class lies on the ambient Lefschetz line $\mathbf{Q}H^2$, whose projection to $H_{\mathrm{prim}}^4(X_{3d}, \mathbf{Q})$ is zero. Primitive Fermat-character spaces lie in this primitive summand [3]. $\square$

Reducedness and irreducibility of W_d are neither asserted nor needed. The conclusion concerns the primitive component of its cohomology class; the total class $2d^2H^2$ is nonzero.

3 A coordinate-power sector obstruction

Let

$$X_m = \left\{ \sum_{i=0}^N x_i^m = 0 \right\} \subset \mathbf{P}_{\mathbf{C}}^N, \quad d \mid m,$$

and let

$$G_m = \mu_m^{N+1} / \mu_m^{\mathrm{diag}}, \quad K_d = \mu_d^{N+1} / \mu_d^{\mathrm{diag}} \subset G_m,$$

where the inclusion is induced by $\mu_d \subset \mu_m$. A character of G_m is represented by $\alpha = (a_0, \dots, a_N) \in (\mathbf{Z}/m)^{N+1}$ with $\sum_i a_i = 0$.

Theorem 3.1. *If a cohomology class v on X_m is invariant under K_d , then its α -character projection is zero unless*

$$d \mid a_i \quad \text{for every } i.$$

In particular this applies to the fundamental class of a cycle defined by homogeneous equations in $x_0^d, \dots, x_N^d$.

Proof. Finite abelian group representations over $\mathbf{C}$ are semisimple. If π_α denotes the character projector, averaging a K_d -invariant class first over K_d shows

$$\pi_\alpha(v) = 0 \quad \text{unless} \quad \chi_\alpha|_{K_d} = 1.$$

Explicitly,

$$\chi_\alpha|_{K_d}[(\zeta_i)] = \prod_i \zeta_i^{a_i}.$$

This is well defined on the diagonal quotient because $\sum_i a_i = 0 \pmod{m}$. Setting all but one coordinate equal to 1 shows that the restriction is trivial exactly when every a_i is divisible by d . Equations in the x_i^d are fixed by K_d , so their scheme and fundamental cycle are invariant. $\square$

For the fourfold X_{3d} , a primitive $(2, 2)$ character surviving this test must be a permutation of

$$(d, d, d, 2d, 2d, 2d).$$

Indeed, primitive Fermat characters have all six coordinates nonzero, and the $(2, 2)$ condition is

$$\sum_i \langle a_i \rangle = 3(3d) = 9d$$

for least-positive representatives [3]. Divisibility by d makes each coordinate d or $2d$. If r coordinates are $2d$, their sum is $(6 + r)d$, so $r = 3$.

4 Degree 33 and the scope of the answer

For $d = 11$ the complete-intersection computation gives

$$[W_{11}] = 242H^2, \quad \deg W_{11} = 7986.$$

Da Silva's highlighted exceptional character is

$$\alpha = (7, 10, 13, 19, 22, 28) \pmod{33}.$$

It is not coordinatewise divisible by 11, so Theorem 3.1 independently forces its W_{11} -projection to vanish. Corollary 2.2 is stronger: every primitive projection of W_{11} vanishes.

Thus Question 1 of [1] has a negative answer for the proposed surface W . Nothing here proves the exceptional character nonalgebraic, nor does it settle the Hodge conjecture for the degree-33 Fermat fourfold or in general.

The version of record of [1] has the correct term $3e_3$ in Newton's identity. An arXiv version displays $3e_3^3$, which is dimensionally inhomogeneous and is evidently a typographical error. Our argument uses the corrected published formula.

Reproducibility and disclosure

The accompanying dependency-free verifier checks the degree-33 character, the numerical cycle-class specialization, and the symmetry obstruction using exact integer arithmetic. The proof itself is symbolic and does not depend on the program.

This result was found, drafted, and internally audited by an autonomous AI mathematics system. Multiple independent AI agents reconstructed the ideal identity, regular-sequence argument, Chow computation, symmetry obstruction, and source wording; hostile audits specifically tested scheme structure, embedded components, character conventions, and scope. No human endorsement is claimed.

References

- [1] G. da Silva Jr., *Notes on the Hodge conjecture for Fermat varieties*, Experimental Results **2** (2021), e22, doi:10.1017/exp.2021.14.
- [2] W. Fulton, *Intersection Theory*, 2nd ed., Springer, 1998.
- [3] T. Shioda, *The Hodge conjecture for Fermat varieties*, Math. Ann. **245** (1979), 175–184.